5 THE RISING SUN

Once I understood that solar panels were our biggest hope I wanted to understand how they developed. I had just taken a sabbatical after 15 years of Internet consultancy. I wanted something new and this seemed to fit the bill.

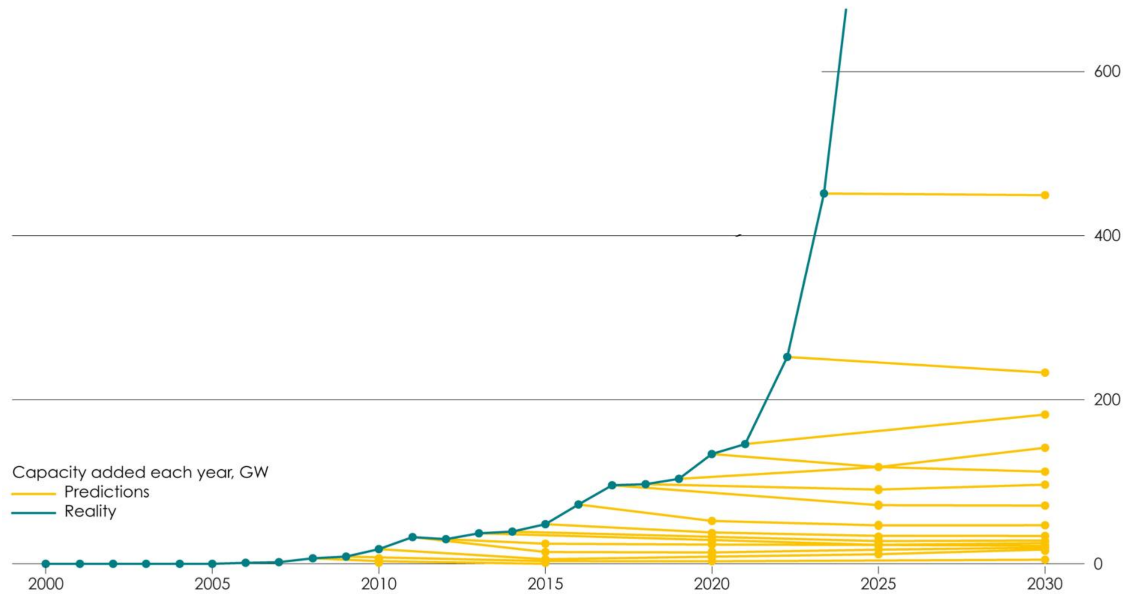
What I found was that solar was still small but had an incredibly predictable yearly exponential growth. For somebody with an Internet background that is a very recognizable phenomenon. So I started interviewing people who worked in the energy industry.

The Dutch fossil fuel company Shell was and is one of the leading energy companies in the world. I interviewed people there and they were not convinced at all. How could *they* miss this when I saw it so clearly. The International Energy Agency (IEA), an enormous organization doing nothing else than collecting energy data and interpreting it for governments, didn't seem that convinced. They had one job. If anybody would know, it would be them I thought naively.

So I took the numbers from their flagship *World Energy Outlook* reports, eager to learn how they thought the exponential revolution would play out. But it didn't. Solar didn't even produce a blip. How could that be?

I dug deeper and made an intuitive leap: "What if they treat solar panels like coal?" Something that is burned in the same year it is produced. Not something that produces energy for 30 years after you put it on your roof like a solar panel does. (Jordan, 2016) (Wiser, 2020) Maybe their model simply can't handle this new dynamic. Their numbers were all in total cumulative solar power production but not in yearly solar panel production.

So I set out to find what these numbers implied about *how many solar panels are sold each year*. The growth of the solar industry if you wish. That information is not shared by the IEA and it took some reverse engineering and guestimating, but it lead to the first version of the graph that would later propel me to some energy system niche fame on twitter and would become the star of articles in *PV Magazine* (Enkhardt, 2018) and *The Economist*. (Economist, 2024) It possibly put some pressure on the IEA. (Hunt, 2020) (Lopez, 2025) What I'm most proud of is its inclusion in a TED talk by Al Gore arguing climate action is unstoppable and "climate realism" is a myth. (Gore, 2025)



The green line going up sharply is the yearly sales of solar panels. A very clear exponential growth. The almost perfectly horizontal yellow lines are the scenarios of the IEA. Year after year, the model of the IEA accepted that the solar industry had grown, but concluded that now this foolishness would end. Not once, but continuously over a period of twenty years, even though people inside the IEA told me my criticism was well known. (Hoekstra, 2017) (Lopez, 2025)

But I didn't go public yet. I felt that first I had to find an explanation before making a fool of myself. And boy, did I find one.

5.1 SIDEBAR: Calculating Exponential Growth

If you want to become an exponential growth detective yourself, to detect what orthodox experts get wrong, you have to know a little bit about exponential growth. If you like making calculations you might find this interesting. But if that's not your hobby, you might want to skip this sidebar.

So what is exponential growth exactly? Let's calculate something first to show how amazing it is. Let's take solar as the example. Over the last 20 years solar grew exponentially with 36% per year on average. (Ember, 2025) What does that mean?

In 2005, solar produced a mere 0.02% of global electricity. (Ember, 2025) But in 2006 it was 36% more. Still only 0.027% or so. But in 2007 again 36% more. And that gets big fast. To get a growth of 36% you multiply by 1.36. And you do that for every year. So after 20 years you have done that 20 times. If you multiply a number by itself 20 times, mathematicians say you raise that number to the power of twenty. When they program it in Excel or Python they use the "^" symbol to signify that. They use the "*" symbol for multiplication.

So how much does 0.02% become after it grows 20 years with 36%?
0.02 * 1.36 ^ 20 = 8.8%.

That's where we are today: solar produces nearly 9% of global electricity, a growth of over 400 times in 20 years! If it kept doing that for another 10 years it would be producing over 190% of our current electricity!

We often use the R squared calculation to determine how well our formula fits the data. R squared ranges from 0% for no fit at all to 100% of a perfect fit. Anything over 90% is very very good, and anything over 95% is truly excellent.

In a spreadsheet program you can put the calculated values next to the empirical values and simply use the RSQ function to compare the two ranges. If I do that and compare my empirical data for solar with my exponential function growing at 36% per year, I find the R squared is 97% which means the fit is exceptionally good.

Now that I know the R squared that expresses how well my exponential growth curve fits the data is 97% it is abundantly clear that solar is growing exponentially and that the growth over the past twenty years was around 36%.